INT315 — Hospital Readmission Prediction with PySpark

This notebook is designed around the INT315 requirement to use PySpark and a Decision Tree classifier. It adds safer preprocessing, meaningful visualizations, threshold tuning, model metrics, and export files for Power BI and the final report.

Dataset: UCI Diabetes 130-US Hospitals for Years 1999–2008
Target: readmission within 30 days (`readmitted == '<30'`)
Important: this is an encounter-level dataset. A random row split can put encounters from the same patient in both train and test sets; we report this limitation and do not claim clinical deployment readiness.

1. Install and import libraries

```
%pip -q install pyspark
```

```
import os
import zipfile
import urllib.request

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from pyspark.sql import SparkSession
from pyspark.sql import functions as F

from pyspark.ml import Pipeline
from pyspark.ml.feature import Imputer, StringIndexer, OneHotEncoder, VectorAssembler
from pyspark.ml.classification import DecisionTreeClassifier, RandomForestClassifier
from pyspark.ml.tuning import CrossValidator, ParamGridBuilder
from pyspark.ml.evaluation import BinaryClassificationEvaluator
from pyspark.ml.evaluation import BinaryClassificationEvaluator
from pyspark.ml.functions import vector_to_array

plt.rcParams["figure.figsize"] = (9, 5)

spark = SparkSession.builder.appName("INT315_Hospital_Readmission").config(
    "spark.sql.shuffle.partitions", "8"
).getOrCreate()

print("Spark version:", spark.version)
```

```
Spark version: 4.0.4
```

2. Download the authentic UCI dataset

The notebook downloads the original UCI ZIP.

```
DATA_DIR = "/content/diabetes_readmission_data"
os.makedirs(DATA_DIR, exist_ok=True)
ZIP_PATH = os.path.join(DATA_DIR, "diabetes.zip")
UCI_ZIP_URL = "https://archive.ics.uci.edu/static/public/296/diabetes+130-us+hospitals+for+years+1999-2008.zip"

if not os.path.exists(os.path.join(DATA_DIR, "diabetic_data.csv")):
    try:
        urllib.request.urlretrieve(UCI_ZIP_URL, ZIP_PATH)
        with zipfile.ZipFile(ZIP_PATH, "r") as zf:
            zf.extractall(DATA_DIR)
    except Exception as e:
        print("Automatic download failed:", repr(e))
        print("Download the ZIP from https://archive.ics.uci.edu/dataset/296/diabetes+130-us-hospitals+for-years+1999-2008")
        print("Upload it to Colab, set ZIP_PATH to the uploaded ZIP path, then rerun this cell.")
        raise

csv_path = os.path.join(DATA_DIR, "diabetic_data.csv")
if not os.path.exists(csv_path):
    candidates = []
    for root, _, files in os.walk(DATA_DIR):
        candidates += [os.path.join(root, f) for f in files if f.lower() == "diabetic_data.csv"]
    if candidates:
        csv_path = candidates[0]
    else:
        raise FileNotFoundError("Could not find diabetic_data.csv in the extracted dataset.")

print("Dataset file:", csv_path)
```

```
Dataset file: /content/diabetes_readmission_data/diabetic_data.csv
```

3. Load and inspect the data

```
raw_df = (
    spark.read
    .option("header", True)
    .option("inferSchema", False)
    .option("quote", '"')
    .option("escape", '"')
    .csv(csv_path)
)
print("Rows:", raw_df.count())
print("Columns:", len(raw_df.columns))
raw_df.printSchema()
raw_df.show(5, truncate=False)
```

```
 |-- gender: string (nullable = true)
 |-- age: string (nullable = true)
 |-- weight: string (nullable = true)
 |-- admission_type_id: string (nullable = true)
 |-- discharge_disposition_id: string (nullable = true)
 |-- admission_source_id: string (nullable = true)
 |-- time_in_hospital: string (nullable = true)
 |-- payer_code: string (nullable = true)
 |-- medical_specialty: string (nullable = true)
 |-- num_lab_procedures: string (nullable = true)
 |-- num_procedures: string (nullable = true)
 |-- num_medications: string (nullable = true)
 |-- number_outpatient: string (nullable = true)
 |-- number_emergency: string (nullable = true)
 |-- number_inpatient: string (nullable = true)
 |-- diag_1: string (nullable = true)
 |-- diag_2: string (nullable = true)
 |-- diag_3: string (nullable = true)
 |-- number_diagnoses: string (nullable = true)
 |-- max_glu_serum: string (nullable = true)
 |-- A1Cresult: string (nullable = true)
 |-- metformin: string (nullable = true)
 |-- repaglinide: string (nullable = true)
 |-- nateglinide: string (nullable = true)
 |-- chlorpropamide: string (nullable = true)
 |-- glimepiride: string (nullable = true)
 |-- acetohexamide: string (nullable = true)
 |-- glipizide: string (nullable = true)
 |-- glyburide: string (nullable = true)
 |-- tolbutamide: string (nullable = true)
 |-- pioglitazone: string (nullable = true)
 |-- rosiglitazone: string (nullable = true)
 |-- acarbose: string (nullable = true)
 |-- miglitol: string (nullable = true)
 |-- troglitazone: string (nullable = true)
 |-- tolazamide: string (nullable = true)
 |-- examide: string (nullable = true)
 |-- citoglipton: string (nullable = true)
 |-- insulin: string (nullable = true)
 |-- glyburide-metformin: string (nullable = true)
 |-- glipizide-metformin: string (nullable = true)
 |-- glimepiride-pioglitazone: string (nullable = true)
 |-- metformin-rosiglitazone: string (nullable = true)
 |-- metformin-pioglitazone: string (nullable = true)
 |-- change: string (nullable = true)
 |-- diabetesMed: string (nullable = true)
 |-- readmitted: string (nullable = true)

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|encounter_id|patient_nbr|race           |gender|age    |weight|admission_type_id|discharge_disposition_id|admission_source_id|time_in_hospital|payer_code|medical_specialty       |num_lab_procedures|num_procedures|num_medications|number_outpatient|number_emergency|number_inpatient|diag_1|diag_2|diag_3|number_diagnoses|max_glu_serum|A1Cresult|metformin|repaglinide|nateglinide|chlorpropamide|glimepiride|acetohexamide|glipizide|glyburide|tolbutamide|pioglitazone|rosiglitazone|acarbose|miglitol|troglitazone|tolazamide|examide|citoglipton|insulin|glyburide-metformin|glipizide-metformin|glimepiride-pioglitazone|metformin-rosiglitazone|metformin-pioglit
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|2278392     |8222157    |Caucasian      |Female|[0-10) |?     |6                |25                      |1                  |1               |?         |Pediatrics-Endocrinology|41                |0             |1              |0                |0               |0               |250.83|?     |?     |1               |None         |None     |No       |No         |No         |No            |No         |No           |No       |No       |No         |No          |No           |No      |No      |No          |No        |No     |No         |No     |No                 |No                 |No                      |No                     |No
|149190      |55629189   |Caucasian      |Female|[10-20)|?     |1                |1                       |7                  |3               |?         |?                       |59                |0             |18             |0                |0               |0               |276   |250.01|255   |9               |None         |None     |No       |No         |No         |No            |No         |No           |No       |No       |No         |No          |No           |No      |No      |No          |No        |No     |No         |Up     |No                 |No                 |No                      |No                     |No
|64410       |86047875   |AfricanAmerican|Female|[20-30)|?     |1                |1                       |7                  |2               |?         |?                       |11                |5             |13             |2                |0               |1               |648   |250   |V27   |6               |None         |None     |No       |No         |No         |No            |No         |No           |Steady   |No       |No         |No          |No           |No      |No      |No          |No        |No     |No         |No     |No                 |No                 |No                      |No                     |No
|500364      |82442376   |Caucasian      |Male  |[30-40)|?     |1                |1                       |7                  |2               |?         |?                       |44                |1             |16             |0                |0               |0               |8     |250.43|403   |7               |None         |None     |No       |No         |No         |No            |No         |No           |No       |No       |No         |No          |No           |No      |No      |No          |No        |No     |No         |Up     |No                 |No                 |No                      |No                     |No
|16680       |42519267   |Caucasian      |Male  |[40-50)|?     |1                |1                       |7                  |1               |?         |?                       |51                |0             |8              |0                |0               |0               |197   |157   |250   |5               |None         |None     |No       |No         |No         |No            |No         |No           |Steady   |No       |No         |No          |No           |No      |No      |No          |No        |No     |No         |Steady |No                 |No                 |No                      |No                     |No
+------------+-----------+---------------+------+-------+------+-----------------+------------------------+-------------------+----------------+----------+------------------------+------------------+--------------+---------------+-----------------+----------------+----------------+------+------+------+----------------+-------------+---------+---------+-----------+-----------+--------------+-----------+-------------+---------+---------+-----------+------------+-------------+--------+--------+------------+----------+-------+-----------+-------+-------------------+-------------------+------------------------+-----------------------+-----------------
only showing top 5 rows
```

4. Inspect original target labels and missing values

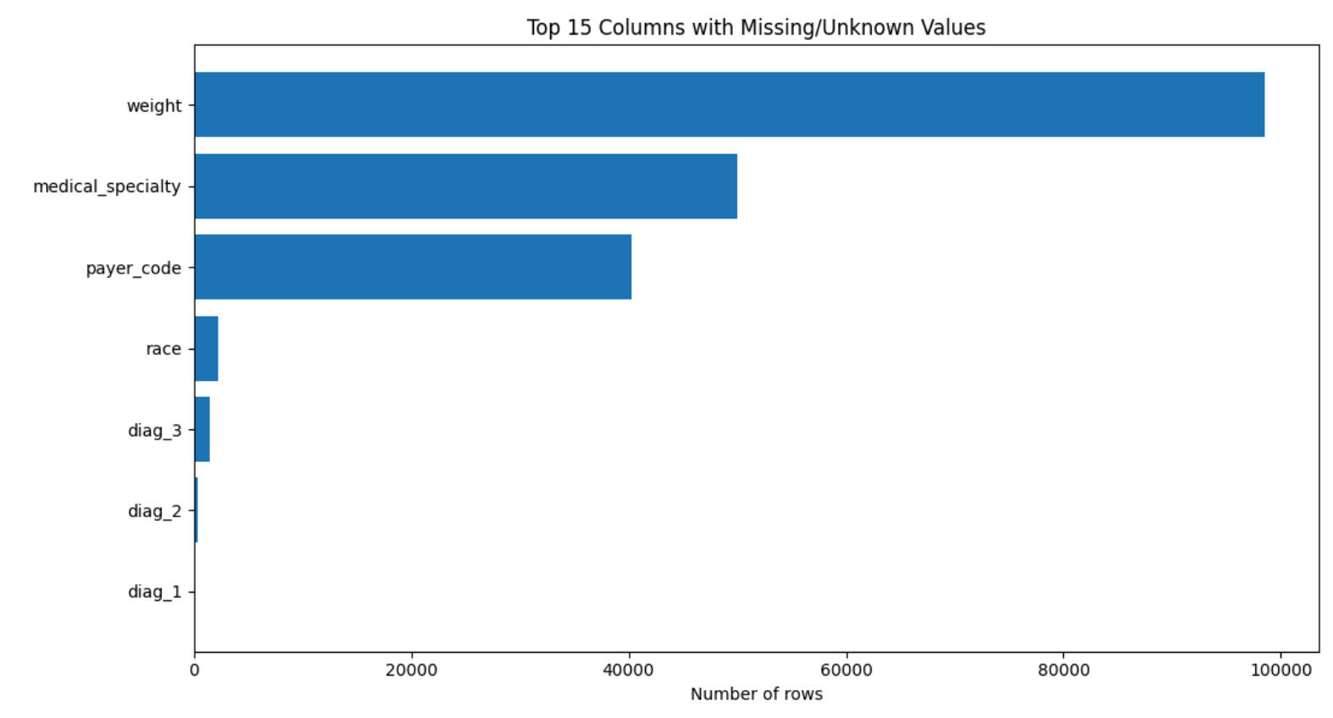
In this dataset, `<30` means readmitted within 30 days, `>30` means readmitted after 30 days, and `NO` means no recorded readmission.
Our binary target marks only `<30` as 1.

```
raw_df.groupBy("readmitted").count().orderBy(F.desc("count")).show()
```

```
+----------+-----+
|readmitted|count|
+----------+-----+
|        NO|54864|
|       >30|35545|
|       <30|11357|
+----------+-----+
```

```
# Count missing or unknown values in each column.
row_count = raw_df.count()
missing_rows = []

for column_name in raw_df.columns:
    missing_count = raw_df.filter(
        F.col(column_name).isNull()
        | (F.trim(F.col(column_name)) == "")
        | (F.col(column_name) == "?")
    ).count()

    if missing_count > 0:
        percentage = round((missing_count / row_count) * 100, 2)
        missing_rows.append((column_name, missing_count, percentage))

# Convert the results into a Pandas DataFrame for display and plotting.
missing_pd = pd.DataFrame(
    missing_rows,
    columns=["Column", "Missing_or_unknown_count", "Percent_of_rows"]
)
missing_pd = missing_pd.sort_values(
    "Missing_or_unknown_count", ascending=False
)

display(missing_pd.head(20))

plt.figure(figsize=(11, 6))
plot_missing = missing_pd.head(15).sort_values("Missing_or_unknown_count")
plt.barh(plot_missing["Column"], plot_missing["Missing_or_unknown_count"])
plt.title("Top 15 Columns with Missing/Unknown Values")
plt.xlabel("Number of rows")
plt.tight_layout()
plt.show()
```

	Column	Missing_or_unknown_count	Percent_of_rows
1	weight	98569	96.86
3	medical_specialty	49949	49.08
2	payer_code	40256	39.56
0	race	2273	2.23
6	diag_3	1423	1.40
5	diag_2	358	0.35
4	diag_1	21	0.02



5. Clean data and define the target

We keep `medical_specialty` so we can analyze it later. It represents physician specialty, **not a confirmed hospital department**. We drop patient/encounter identifiers from model features. `weight` is omitted because it is missing for most encounters.

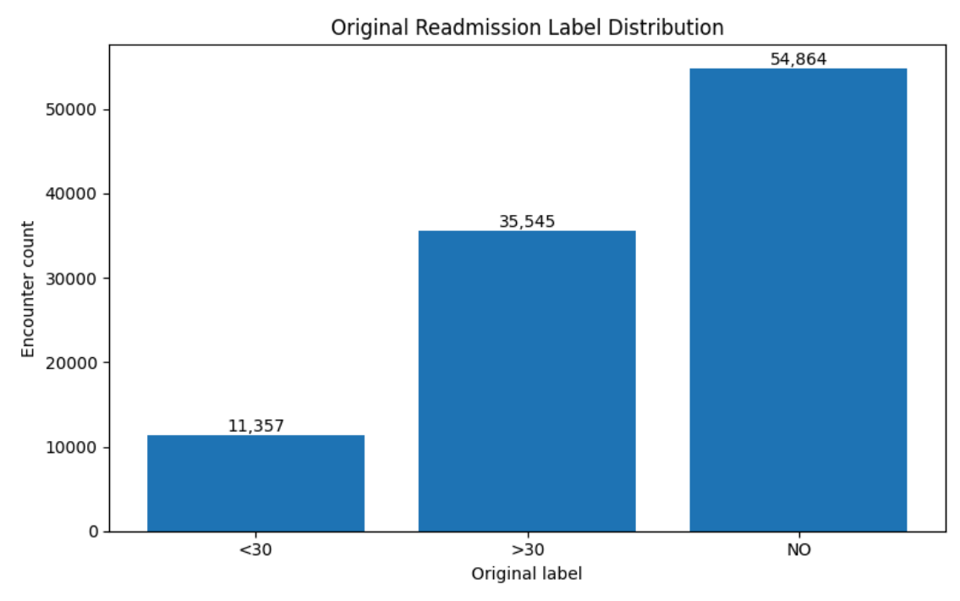
```
from pyspark.sql import functions as F
# Step 1: Check missing values before cleaning
required_fields = [
    "admission_type_id",
    "discharge_disposition_id",
    "admission_source_id"
]
print("Missing values BEFORE cleaning:")

for column_name in required_fields:
    missing_count = raw_df.filter(
        F.col(column_name).isNull() |
        (F.trim(F.col(column_name)) == "") |
        (F.trim(F.col(column_name)) == "?")
    ).count()

    print(f"{column_name}: {missing_count}")

# Step 2: Convert '?' and blank strings to null
clean_df = raw_df

for column_name in clean_df.columns:
    clean_df = clean_df.withColumn(
        column_name,
        F.when(
            F.col(column_name).isNull() |
            (F.trim(F.col(column_name)) == "") |
            (F.trim(F.col(column_name)) == "?"),
            F.lit(None).cast("string")
        ).otherwise(F.trim(F.col(column_name)))
    )

# Step 3: Explicitly replace missing admission/discharge codes
# with "Unknown"
for column_name in required_fields:
    clean_df = clean_df.withColumn(
        column_name,
        F.coalesce(F.col(column_name), F.lit("Unknown"))
    )

# Step 4: Create the 30-day readmission target
clean_df = clean_df.withColumn(
    "readmitted_30_days",
    F.when(F.col("readmitted") == "<30", 1.0).otherwise(0.0)
)

# Step 5: Verify the required fields after cleaning
print("\nMissing values AFTER cleaning:")

for column_name in required_fields:
    remaining = clean_df.filter(
        F.col(column_name).isNull() |
        (F.trim(F.col(column_name)) == "")
    ).count()

    print(f"{column_name}: {remaining}")

print("\nTarget distribution:")
clean_df.groupBy("readmitted_30_days").count().show()

print("Total rows:", clean_df.count())
```

```
Missing values BEFORE cleaning:
admission_type_id: 0
discharge_disposition_id: 0
admission_source_id: 0

Missing values AFTER cleaning:
admission_type_id: 0
discharge_disposition_id: 0
admission_source_id: 0

Target distribution:
+------------------+-----+
|readmitted_30_days|count|
+------------------+-----+
|               1.0|11357|
|               0.0|90409|
+------------------+-----+

Total rows: 101766
```

6. Exploratory graphs

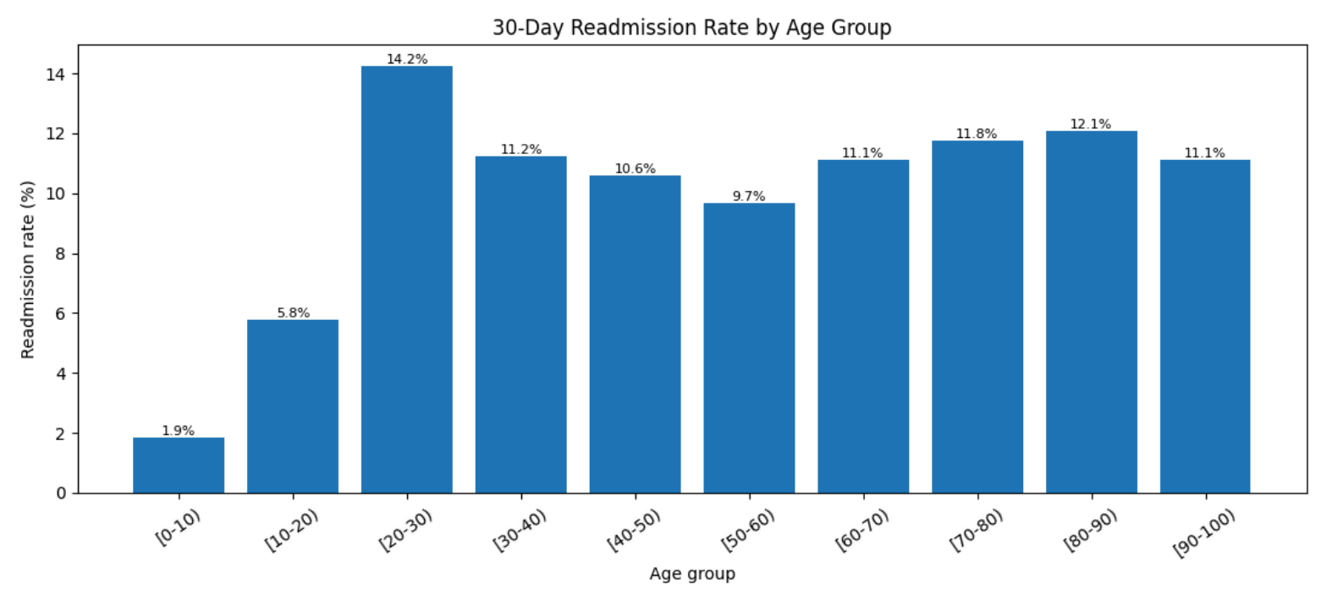
The aggregations are performed in Spark; only the small summary tables are collected into Pandas for plotting. This avoids collecting the full patient-level dataset into local memory.

```
target_counts = (
    clean_df.groupBy("readmitted")
    .count().orderBy("readmitted")
    .toPandas()
)
plt.figure(figsize=(8, 5))
bars = plt.bar(target_counts["readmitted"].astype(str), target_counts["count"])
plt.title("Original Readmission Label Distribution")
plt.xlabel("Original label")
plt.ylabel("Encounter count")
for b in bars:
    plt.text(b.get_x()+b.get_width()/2, b.get_height(), f"{int(b.get_height()):,}", ha="center", va="bottom")
plt.tight_layout()
plt.show()
```



```
age_summary = (
    clean_df.groupBy("age")
    .agg(
        F.count("*").alias("total_encounters"),
        F.sum("readmitted_30_days").alias("readmitted_within_30_days")
    )
    .withColumn("readmission_rate_pct", F.round(100*F.col("readmitted_within_30_days")/F.col("total_encounters"), 2))
)
age_order = ["[0-10)", "[10-20)", "[20-30)", "[30-40)", "[40-50)", "[50-60)", "[60-70)", "[70-80)", "[80-90)", "[90-100)"]
age_pd = age_summary.toPandas()
age_pd["age"] = pd.Categorical(age_pd["age"], categories=age_order, ordered=True)
age_pd = age_pd.sort_values("age")
display(age_pd)

plt.figure(figsize=(11, 5))
bars = plt.bar(age_pd["age"].astype(str), age_pd["readmission_rate_pct"])
plt.title("30-Day Readmission Rate by Age Group")
plt.xlabel("Age group")
plt.ylabel("Readmission rate (%)")
plt.xticks(rotation=35)
for b in bars:
    plt.text(b.get_x()+b.get_width()/2, b.get_height(), f"{b.get_height():.1f}%", ha="center", va="bottom", fontsize=8)
plt.tight_layout()
plt.show()
```

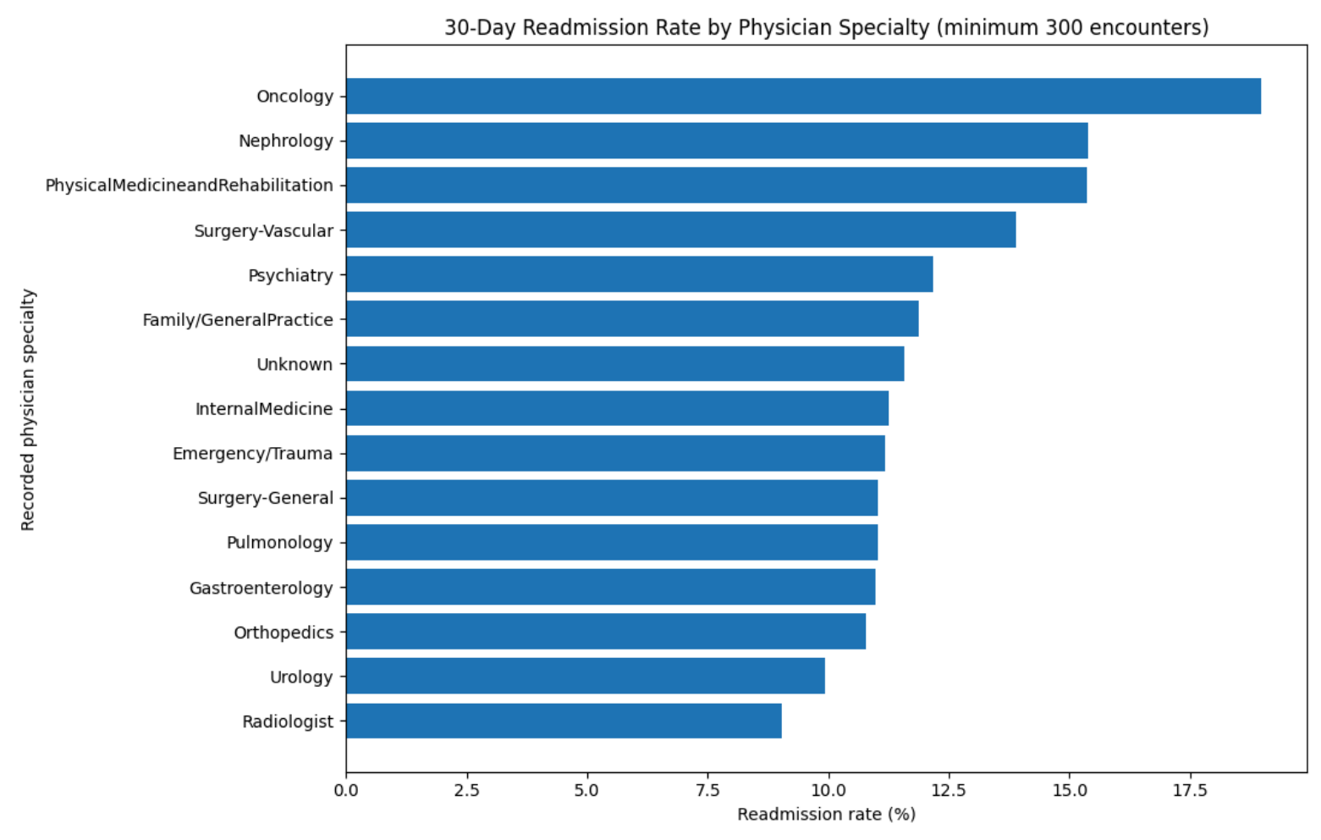
	age	total_encounters	readmitted_within_30_days	readmission_rate_pct
6	[0-10)	161	3.0	1.86
3	[10-20)	691	40.0	5.79
1	[20-30)	1657	236.0	14.24
5	[30-40)	3775	424.0	11.23
4	[40-50)	9685	1027.0	10.60
8	[50-60)	17256	1668.0	9.67
7	[60-70)	22483	2502.0	11.13
9	[70-80)	26068	3069.0	11.77
2	[80-90)	17197	2078.0	12.08
0	[90-100)	2793	310.0	11.10



Next steps: Generate code with `age_pd` New interactive sheet

```
# Physician specialty is used as a proxy for department because the dataset does not provide a clean department field.
specialty_summary = (
    clean_df.groupBy("medical_specialty")
    .agg(F.count("*").alias("total_encounters"),
         F.sum("readmitted_30_days").alias("readmitted_within_30_days"))
    .withColumn("readmission_rate_pct", F.round(100*F.col("readmitted_within_30_days")/F.col("total_encounters"), 2))
    .filter(F.col("total_encounters") >= 300)
    .orderBy(F.desc("readmission_rate_pct"))
)
specialty_pd = specialty_summary.limit(15).toPandas()
display(specialty_pd)

plt.figure(figsize=(11, 7))
specialty_plot = specialty_pd.sort_values("readmission_rate_pct")
plt.barh(specialty_plot["medical_specialty"].fillna("Unknown"), specialty_plot["readmission_rate_pct"])
plt.title("30-Day Readmission Rate by Physician Specialty (minimum 300 encounters)")
plt.xlabel("Readmission rate (%)")
plt.ylabel("Recorded physician specialty")
plt.tight_layout()
plt.show()
```

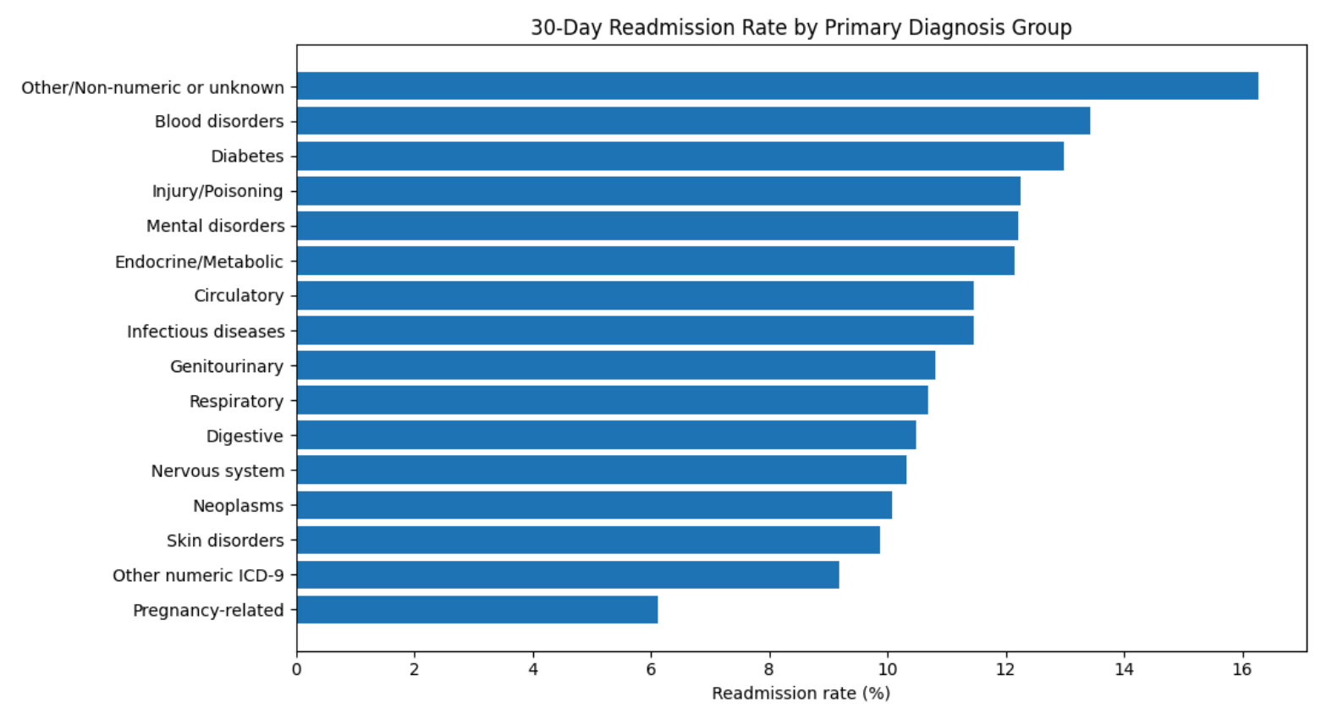
	medical_specialty	total_encounters	readmitted_within_30_days	readmission_rate_pct
0	Oncology	348	66.0	18.97
1	Nephrology	1613	248.0	15.38
2	PhysicalMedicineandRehabilitation	391	60.0	15.35
3	Surgery-Vascular	533	74.0	13.88
4	Psychiatry	854	104.0	12.18
5	Family/GeneralPractice	7440	883.0	11.87
6	None	49949	5781.0	11.57
7	InternalMedicine	14635	1646.0	11.25
8	Emergency/Trauma	7565	846.0	11.18
9	Surgery-General	3099	342.0	11.04
10	Pulmonology	871	96.0	11.02
11	Gastroenterology	564	62.0	10.99
12	Orthopedics	1400	151.0	10.79
13	Urology	685	68.0	9.93
14	Radiologist	1140	103.0	9.04



Next steps: Generate code with `specialty_pd` New interactive sheet

```
# Group primary diagnoses into broad ICD-9 ranges for an interpretable summary.
diag = clean_df.withColumn("diag1_text", F.trim(F.col("diag_1")))
diag = diag.withColumn("diag1_num", F.expr("try_cast(NULLIF(regexp_extract(diag1_text, '^(\\\\d+)', 1), '') AS INT)"))
diag = diag.withColumn(
    "diagnosis_group",
    F.when(F.col("diag1_num") == 250, "Diabetes")
     .when(F.col("diag1_num").between(390,459), "Circulatory")
     .when(F.col("diag1_num").between(460,519), "Respiratory")
     .when(F.col("diag1_num").between(520,579), "Digestive")
     .when(F.col("diag1_num").between(800,999), "Injury/Poisoning")
     .when(F.col("diag1_num").between(580,629), "Genitourinary")
     .when(F.col("diag1_num").between(140,239), "Neoplasms")
     .when(F.col("diag1_num").between(240,279), "Endocrine/Metabolic")
     .when(F.col("diag1_num").between(280,289), "Blood disorders")
     .when(F.col("diag1_num").between(290,319), "Mental disorders")
     .when(F.col("diag1_num").between(320,389), "Nervous system")
     .when(F.col("diag1_num").between(680,709), "Skin disorders")
     .when(F.col("diag1_num").between(1,139), "Infectious diseases")
     .when(F.col("diag1_num").between(630,679), "Pregnancy-related")
     .when(F.col("diag1_num").isNotNull(), "Other numeric ICD-9")
     .otherwise("Other/Non-numeric or unknown")
)
diagnosis_summary = (
    diag.groupBy("diagnosis_group")
    .agg(F.count("*").alias("total_encounters"),
         F.sum("readmitted_30_days").alias("readmitted_within_30_days"))
    .withColumn("readmission_rate_pct", F.round(100*F.col("readmitted_within_30_days")/F.col("total_encounters"), 2))
    .filter(F.col("total_encounters") >= 100)
    .orderBy(F.desc("readmission_rate_pct"))
)
diagnosis_pd = diagnosis_summary.toPandas()
display(diagnosis_pd)

plt.figure(figsize=(11, 6))
diagnosis_plot = diagnosis_pd.sort_values("readmission_rate_pct")
plt.barh(diagnosis_plot["diagnosis_group"], diagnosis_plot["readmission_rate_pct"])
plt.title("30-Day Readmission Rate by Primary Diagnosis Group")
plt.xlabel("Readmission rate (%)")
plt.tight_layout()
plt.show()
```

	diagnosis_group	total_encounters	readmitted_within_30_days	readmission_rate_pct
0	Other/Non-numeric or unknown	1666	271.0	16.27
1	Blood disorders	1103	148.0	13.42
2	Diabetes	8757	1137.0	12.98
3	Injury/Poisoning	6974	854.0	12.25
4	Mental disorders	2262	276.0	12.20
5	Endocrine/Metabolic	2702	328.0	12.14
6	Circulatory	30336	3474.0	11.45
7	Infectious diseases	2768	317.0	11.45
8	Genitourinary	5078	549.0	10.81
9	Respiratory	10407	1112.0	10.69
10	Digestive	9208	966.0	10.49
11	Nervous system	1211	125.0	10.32
12	Neoplasms	3433	346.0	10.08
13	Skin disorders	2530	250.0	9.88
14	Other numeric ICD-9	12644	1162.0	9.19
15	Pregnancy-related	687	42.0	6.11



Next steps: Generate code with `diagnosis_pd` New interactive sheet

7. Prepare features carefully

The three fields `admission_type_id`, `discharge_disposition_id`, and `admission_source_id` are coded categories, not continuous measurements, so they are handled as categorical features.

Numeric measurement/count fields are explicitly cast to double. Categorical missing values become `"Unknown"`. The ID columns are excluded from predictors to reduce direct identifier leakage.

Preparing columns for the model

We separate numerical and categorical columns. Numerical columns contain values such as number of medicines; categorical columns contain labels such as gender or medical specialty. IDs and the answer column are excluded from the predictors.

```
excluded_cols = [
    "encounter_id", "patient_nbr", "readmitted", "readmitted_30_days",
    "weight", "diag1_text", "diag1_num", "diagnosis_group"
]

numeric_cols = [
    "time_in_hospital", "num_lab_procedures", "num_procedures",
    "num_medications", "number_outpatient", "number_emergency",
    "number_inpatient", "number_diagnoses"
]

all_predictor_cols = []
for column_name in clean_df.columns:
    if column_name not in excluded_cols:
        all_predictor_cols.append(column_name)

categorical_cols = []
for column_name in all_predictor_cols:
    if column_name not in numeric_cols:
        categorical_cols.append(column_name)

coded_categorical_cols = [
    "admission_type_id", "discharge_disposition_id", "admission_source_id"
]

model_df = clean_df.select(
    *[
        F.col(column_name).cast("double").alias(column_name)
        if column_name in numeric_cols
        else F.coalesce(
            F.col(column_name).cast("string"), F.lit("Unknown")
        ).alias(column_name)
        for column_name in numeric_cols + categorical_cols
    ],
    F.col("readmitted_30_days").cast("double").alias("label")
)

for column_name in numeric_cols:
    model_df = model_df.withColumn(
        column_name,
        F.when(F.isnan(F.col(column_name)), None).otherwise(F.col(column_name))
    )

print("Numeric predictors:", numeric_cols)
print("Categorical predictor count:", len(categorical_cols))
print("Coded categorical IDs:", coded_categorical_cols)
print("Model rows:", model_df.count())
```

```
Numeric predictors: ['time_in_hospital', 'num_lab_procedures', 'num_procedures', 'num_medications', 'number_outpatient', 'number_emergency', 'number_inpatient', 'number_diagnoses']
Categorical predictor count: 38
Coded categorical IDs: ['admission_type_id', 'discharge_disposition_id', 'admission_source_id']
Model rows: 101766
```

8. Train/validation/test split and leakage-safe preprocessing

```
train_all, test_df = model_df.randomSplit([0.8, 0.2], seed=42)
fit_df, val_df = train_all.randomSplit([0.8, 0.2], seed=123)
print("Fit rows:", fit_df.count())
print("Validation rows:", val_df.count())
print("Test rows:", test_df.count())
```

```
Fit rows: 65104
Validation rows: 16287
Test rows: 20375
```

Converting data into a format the model can use

The Decision Tree needs one feature vector. The Imputer fills missing numbers, StringIndexer converts text categories to numbers, OneHotEncoder represents categories safely, and VectorAssembler combines everything into one features column. Class weights help the model pay more attention to the smaller readmission group.

```
imputer = Imputer(
    inputCols=numeric_cols,
    outputCols=[c + "_imputed" for c in numeric_cols],
    strategy="median"
)
indexers = [
    StringIndexer(inputCol=c, outputCol=c+"_idx", handleInvalid="keep")
    for c in categorical_cols
]
encoder = OneHotEncoder(
    inputCols=[c+"_idx" for c in categorical_cols],
    outputCols=[c+"_ohe" for c in categorical_cols],
    handleInvalid="keep"
)
assembler_inputs = [c+"_imputed" for c in numeric_cols] + [c+"_ohe" for c in categorical_cols]
assembler = VectorAssembler(inputCols=assembler_inputs, outputCol="features", handleInvalid="keep")
# Handle imbalance with per-row weights for the Decision Tree.
class_counts = {r["label"]: r["count"] for r in fit_df.groupBy("label").count().collect()}
majority = max(class_counts.values())
minority = min(class_counts.values())
positive_weight = majority / max(minority, 1)
fit_weighted = fit_df.withColumn("class_weight", F.when(F.col("label") == 1.0, F.lit(float(positive_weight))).otherwise(F.lit(1.0)))
print("Training class counts:", class_counts)
print("Positive class weight:", round(positive_weight, 3))
```

```
Training class counts: {1.0: 7251, 0.0: 57853}
Positive class weight: 7.979
```

```
# 1. Evaluator on probability (ensures proper ROC-AUC calculation)
evaluator = BinaryClassificationEvaluator(
    labelCol="label",
    rawPredictionCol="probability",
    metricName="areaUnderROC"
)
dt = DecisionTreeClassifier(
    labelCol="label",
    featuresCol="features",
    weightCol="class_weight",
    maxBins=64,
    seed=42
)
cv_pipeline = Pipeline(stages=[imputer] + indexers + [encoder, assembler, dt])
paramGrid = (ParamGridBuilder()
             .addGrid(dt.maxDepth, [4, 6, 8])
             .build())

#5-Fold Cross-Validator
cv = CrossValidator(
    estimator=cv_pipeline,
    estimatorParamMaps=paramGrid,
    evaluator=evaluator,
    numFolds=5,
    seed=42
)

#Fit and print results
print("Running 5-Fold Cross-Validation...")
cv_model = cv.fit(fit_weighted)

print("\n5-Fold Cross-Validation Results")
print("------------------------------")
print(f"Max Depth 4: ROC-AUC = {cv_model.avgMetrics[0]:.4f}")
print(f"Max Depth 6: ROC-AUC = {cv_model.avgMetrics[1]:.4f}")
print(f"Max Depth 8: ROC-AUC = {cv_model.avgMetrics[2]:.4f}")
print(f"\nBest Max Depth: {[4, 6, 8][cv_model.avgMetrics.index(max(cv_model.avgMetrics))]}")
print(f"Best Cross-Validation ROC-AUC: {max(cv_model.avgMetrics):.4f}")
print("\n5-Fold Cross-Validation Completed Successfully!")
print("Best Decision Tree model selected.")
dt_model = cv_model.bestModel
```

```
Running 5-Fold Cross-Validation...

5-Fold Cross-Validation Results
------------------------------
Max Depth 4: ROC-AUC = 0.6440
Max Depth 6: ROC-AUC = 0.6517
Max Depth 8: ROC-AUC = 0.6484

Best Max Depth: 6
Best Cross-Validation ROC-AUC: 0.6517

5-Fold Cross-Validation Completed Successfully!
Best Decision Tree model selected.
```

9. Tune the decision threshold on validation data

Spark's default threshold is 0.5, which can miss many positive cases in an imbalanced dataset. We select a threshold on the validation set using F1, then evaluate that fixed threshold on the held-out test set. The test set is not used to choose the threshold.

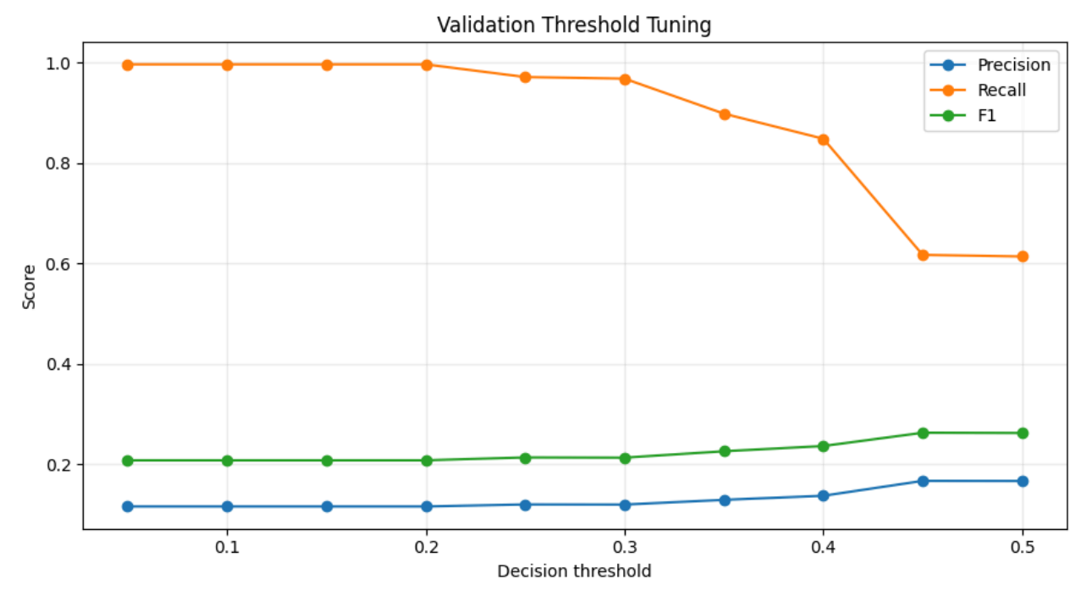
Choosing a classification threshold

The model produces a probability. We test the same thresholds from 0.05 to 0.50 and choose the threshold with the best validation F1 score (using recall as the tie-breaker).

```
def add_positive_probability(pred_df):
    return pred_df.withColumn("prob_readmit", vector_to_array(F.col("probability"))[1])

val_pred = add_positive_probability(dt_model.transform(val_df))
threshold_rows = []
for threshold in [0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50]:
    temp = val_pred.withColumn("threshold_prediction", (F.col("prob_readmit") >= F.lit(threshold)).cast("double"))
    counts = temp.agg(
        F.sum(F.when((F.col("label")==1)&(F.col("threshold_prediction")==1),1).otherwise(0)).alias("tp"),
        F.sum(F.when((F.col("label")==0)&(F.col("threshold_prediction")==1),1).otherwise(0)).alias("fp"),
        F.sum(F.when((F.col("label")==1)&(F.col("threshold_prediction")==0),1).otherwise(0)).alias("fn")
    ).first()
    tp, fp, fn = [int(counts[k] or 0) for k in ["tp","fp","fn"]]
    precision = tp/(tp+fp) if tp+fp else 0.0
    recall = tp/(tp+fn) if tp+fn else 0.0
    f1 = 2*precision*recall/(precision+recall) if precision+recall else 0.0
    threshold_rows.append((threshold, precision, recall, f1))
threshold_pd = pd.DataFrame(threshold_rows, columns=["threshold","precision","recall","f1"])
display(threshold_pd)
best_threshold = float(threshold_pd.sort_values(["f1","recall"], ascending=False).iloc[0]["threshold"])
print("Selected validation threshold:", best_threshold)

plt.figure(figsize=(9,5))
plt.plot(threshold_pd["threshold"], threshold_pd["precision"], marker="o", label="Precision")
plt.plot(threshold_pd["threshold"], threshold_pd["recall"], marker="o", label="Recall")
plt.plot(threshold_pd["threshold"], threshold_pd["f1"], marker="o", label="F1")
plt.xlabel("Decision threshold")
plt.ylabel("Score")
plt.title("Validation Threshold Tuning")
plt.legend()
plt.grid(alpha=0.25)
plt.tight_layout()
plt.show()
```

	threshold	precision	recall	f1
0	0.05	0.116113	0.996245	0.207986
1	0.10	0.116113	0.996245	0.207986
2	0.15	0.116113	0.996245	0.207986
3	0.20	0.116113	0.996245	0.207986
4	0.25	0.120050	0.971030	0.213683
5	0.30	0.119843	0.967811	0.213277
6	0.35	0.129246	0.898069	0.225972
7	0.40	0.137335	0.848176	0.236394
8	0.45	0.167006	0.616953	0.262857
9	0.50	0.166788	0.613734	0.262295

```
Selected validation threshold: 0.45
```



Next steps: Generate code with `threshold_pd` New interactive sheet

10. Final test evaluation

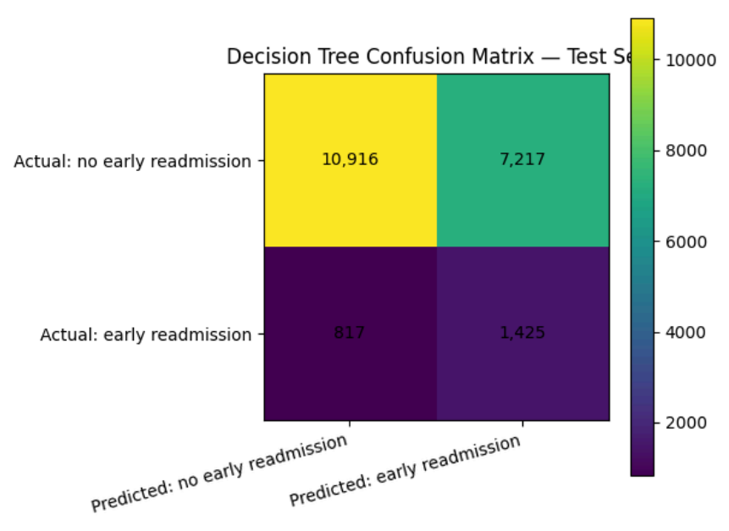
Metrics below are calculated on the held-out test set. ROC-AUC uses the model's probability vector, not the hard class prediction. Accuracy is reported but should not be the only model-selection criterion.

```
test_pred = add_positive_probability(dt_model.transform(test_df))
test_pred = test_pred.withColumn("tuned_prediction", (F.col("prob_readmit") >= F.lit(best_threshold)).cast("double"))

def binary_counts(pred, prediction_col):
    row = pred.agg(
        F.sum(F.when((F.col("label")==1)&(F.col(prediction_col)==1),1).otherwise(0)).alias("tp"),
        F.sum(F.when((F.col("label")==0)&(F.col(prediction_col)==1),1).otherwise(0)).alias("fp"),
        F.sum(F.when((F.col("label")==1)&(F.col(prediction_col)==0),1).otherwise(0)).alias("fn"),
        F.sum(F.when((F.col("label")==0)&(F.col(prediction_col)==0),1).otherwise(0)).alias("tn")
    ).first()
    return {k:int(row[k] or 0) for k in ["tp","fp","fn","tn"]}

counts = binary_counts(test_pred, "tuned_prediction")
tp, fp, fn, tn = counts["tp"], counts["fp"], counts["fn"], counts["tn"]
accuracy = (tp+tn)/(tp+tn+fp+fn) if tp+tn+fp+fn else 0
precision = tp/(tp+fp) if tp+fp else 0
recall = tp/(tp+fn) if tp+fn else 0
f1 = 2*precision*recall/(precision+recall) if precision+recall else 0

auc_eval = BinaryClassificationEvaluator(
    labelCol="label",
    rawPredictionCol="probability",
    metricName="areaUnderROC"
)
roc_auc = auc_eval.evaluate(test_pred)

metrics_pd = pd.DataFrame([{
    "Model":"Weighted PySpark Decision Tree",
    "Threshold":best_threshold,
    "Accuracy":accuracy,
    "Precision_positive":precision,
    "Recall_positive":recall,
    "F1_positive":f1,
    "ROC_AUC":roc_auc,
    "TP":tp,"FP":fp,"FN":fn,"TN":tn
}])
display(metrics_pd)
print("Confusion counts:", counts)
```

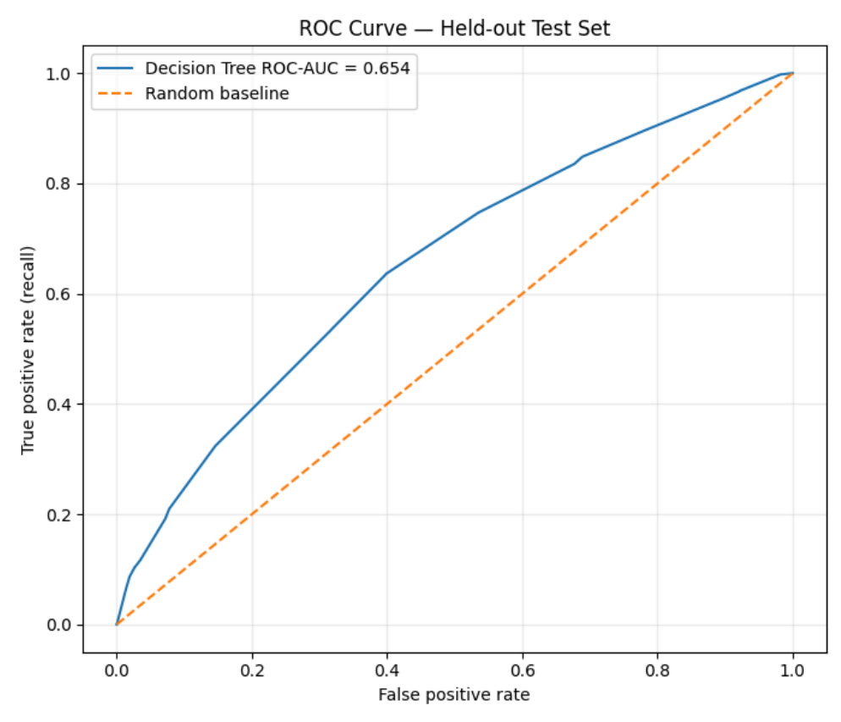
	Model	Threshold	Accuracy	Precision_positive	Recall_positive	F1_positive	ROC_AUC	TP	FP	FN	TN
0	Weighted PySpark Decision Tree	0.45	0.605693	0.164892	0.635593	0.261852	0.654001	1425	7217	817	10916

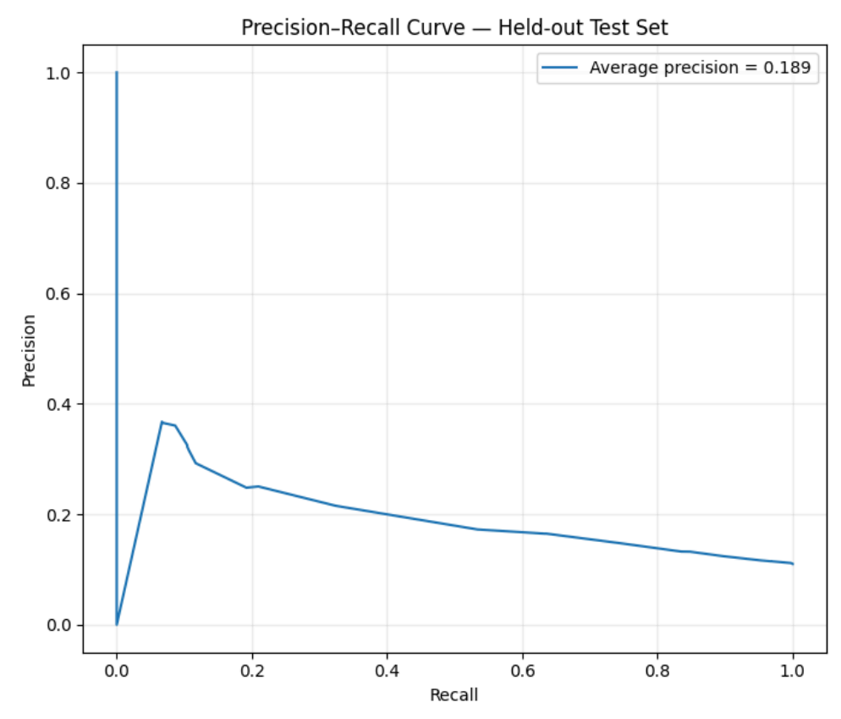
```
Confusion counts: {'tp': 1425, 'fp': 7217, 'fn': 817, 'tn': 10916}
```

```
# Confusion matrix plotted from actual final test counts
cm = np.array([[tn, fp], [fn, tp]])
fig, ax = plt.subplots(figsize=(6,5))
im = ax.imshow(cm)
ax.set_xticks([0,1], labels=["Predicted: no early readmission","Predicted: early readmission"], rotation=15, ha="right")
ax.set_yticks([0,1], labels=["Actual: no early readmission","Actual: early readmission"])
ax.set_title("Decision Tree Confusion Matrix — Test Set")
for i in range(2):
    for j in range(2):
        ax.text(j, i, f"{cm[i,j]:,}", ha="center", va="center")
fig.colorbar(im, ax=ax)
plt.tight_layout()
plt.show()
```



```
# ROC curve from Spark's scored test predictions.
# Collect only the small (label, probability) columns needed for plotting.
roc_pd = test_pred.select("label", "prob_readmit").toPandas()
from sklearn.metrics import roc_curve, roc_auc_score, precision_recall_curve, average_precision_score
fpr, tpr, _ = roc_curve(roc_pd["label"], roc_pd["prob_readmit"])
auc_from_prob = roc_auc_score(roc_pd["label"], roc_pd["prob_readmit"])

plt.figure(figsize=(7,6))
plt.plot(fpr, tpr, label=f"Decision Tree ROC-AUC = {auc_from_prob:.3f}")
plt.plot([0,1], [0,1], linestyle="--", label="Random baseline")
plt.xlabel("False positive rate")
plt.ylabel("True positive rate (recall)")
plt.title("ROC Curve — Held-out Test Set")
plt.legend()
plt.grid(alpha=0.25)
plt.tight_layout()
plt.show()

prec_curve, rec_curve, _ = precision_recall_curve(roc_pd["label"], roc_pd["prob_readmit"])
ap = average_precision_score(roc_pd["label"], roc_pd["prob_readmit"])
plt.figure(figsize=(7,6))
plt.plot(rec_curve, prec_curve, label=f"Average precision = {ap:.3f}")
plt.xlabel("Recall")
plt.ylabel("Precision")
plt.title("Precision-Recall Curve — Held-out Test Set")
plt.legend()
plt.grid(alpha=0.25)
plt.tight_layout()
plt.show()
print("ROC-AUC recomputed from probability:", round(float(auc_from_prob), 4))
print("Average precision (PR-AUC summary):", round(float(ap), 4))
```





```
ROC-AUC recomputed from probability: 0.654
Average precision (PR-AUC summary): 0.1895
```

11. Optional benchmark: PySpark Random Forest

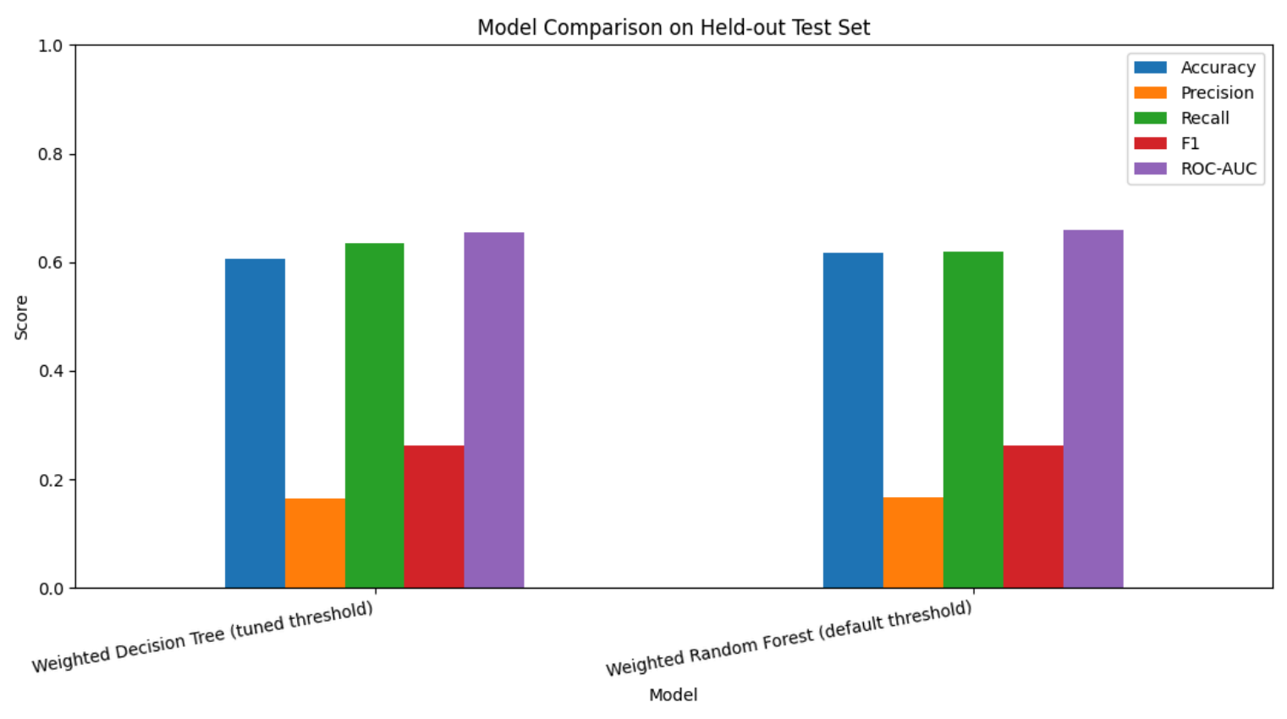
This is an optional comparison model. It uses the same feature pipeline and split. Its output is not automatically “better”; compare recall, precision, F1 and ROC-AUC as well as accuracy. If runtime is tight, skip this section and keep the Decision Tree as the required model.

Comparing with Random Forest

This final section trains the comparison model using the same prepared features and training data, then compares its test results with the Decision Tree.

```
# Reuse the preprocessing stages but replace the classifier.
rf = RandomForestClassifier(
    labelCol="label", featuresCol="features", weightCol="class_weight",
    numTrees=40, maxDepth=8, seed=42
)
rf_pipeline = Pipeline(stages=[imputer] + indexers + [encoder, assembler, rf])
rf_model = rf_pipeline.fit(fit_weighted)
rf_test_pred = add_positive_probability(rf_model.transform(test_df))
rf_auc = BinaryClassificationEvaluator(
    labelCol="label",
    rawPredictionCol="probability",
    metricName="areaUnderROC"
).evaluate(rf_test_pred)

rf_test_default = rf_test_pred.withColumn("predicted_class", F.col("prediction").cast("double"))
rf_counts = binary_counts(rf_test_default, "predicted_class")
rf_tp, rf_fp, rf_fn, rf_tn = rf_counts["tp"], rf_counts["fp"], rf_counts["fn"], rf_counts["tn"]
rf_acc = (rf_tp+rf_tn)/(rf_tp+rf_tn+rf_fp+rf_fn)
rf_prec = rf_tp/(rf_tp+rf_fp) if rf_tp+rf_fp else 0
rf_rec = rf_tp/(rf_tp+rf_fn) if rf_tp+rf_fn else 0
rf_f1 = 2*rf_prec*rf_rec/(rf_prec+rf_rec) if rf_prec+rf_rec else 0

comparison_pd = pd.DataFrame([
    {"Model":"Weighted Decision Tree (tuned threshold)", "Accuracy":accuracy, "Precision":precision, "Recall":recall, "F1":f1, "ROC-AUC":roc_auc},
    {"Model":"Weighted Random Forest (default threshold)", "Accuracy":rf_acc, "Precision":rf_prec, "Recall":rf_rec, "F1":rf_f1, "ROC-AUC":rf_auc}
])
display(comparison_pd)

comparison_pd.set_index("Model")[["Accuracy","Precision","Recall","F1","ROC-AUC"]].plot(kind="bar", figsize=(11,6))
plt.title("Model Comparison on Held-out Test Set")
plt.ylabel("Score")
plt.xticks(rotation=10, ha="right")
plt.ylim(0,1)
plt.tight_layout()
plt.show()
```

	Model	Accuracy	Precision	Recall	F1	ROC-AUC
0	Weighted Decision Tree (tuned threshold)	0.605693	0.164892	0.635593	0.261852	0.654001
1	Weighted Random Forest (default threshold)	0.617178	0.166467	0.618644	0.262342	0.658278



Next steps: Generate code with comparison_pd New interactive sheet